

CSC 345 — Exam 1 Study Guide

Start Here

Guide updated: 2026-09-24 afternoon (America/Phoenix) -- Start Here rewrite. Reviewed 2026-09-24 evening by Claude Opus 5.5 (see §8c).

If you only read one thing: submit HW2 in the late window (by **Sat Sep 26 2:00 PM**) if it isn't in. The one early session to protect is **Sat Sep 26 after 2:00 PM: Topic 2 proofs**, because proof completeness was the HW1 signal. The core exam block is Fri Oct 2 - Mon Oct 5. Exam day is **back-to-back after the 335 midterm**, so there is no study time between them.

Next scheduled assessment

Field	Value	Label
Assessment	CSC 345 Exam 1	confirmed item
Date	Thu Oct 8, 2026	confirmed (professorlynam syllabus schedule)
Time	All-day calendar; class meets 2:00-3:15 PM	clock NV
Format	In person (syllabus); Gradescope role unknown	format NV
Coverage	Topics 1-3 (Review DS; Algorithm Analysis; Graphs)	PROVISIONAL -- schedule-based review
Topic 3 depth	Weeks 5-6 (week 6 = Sep 29-Oct 1). Slides include Dijkstra, MCST, topo sort	how far lecture gets before Oct 8 is unverified
Topic 4	Internal Sort -- slides not linked	excluded
Same day	CSC 335 Midterm 12:30-1:45 same room	plan energy

Do **not** schedule study during class blocks (Tue/Thu through Dec 9, Arizona time): CSC 252 9:30-10:45 AM; CSC 335 12:30-1:45 PM; CSC 345 2:00-3:15 PM; CSC 337 3:30-4:45 PM; CSC 380 5:00-6:15 PM. All times below are **Arizona (MST, UTC-7)**. Combined daily order: `weekly/2026-09-24-weekly-study-order.md`.

Tonight / late window: CSC 345 **HW2** primary due was **Thu Sep 24 2:00 PM**. Tracker still showed No Submission at 14:03. **Late window through Sat Sep 26 2:00 PM**. That outranks Exam 1 study until it is submitted. PP2 due Tue Oct 13 2:00 PM (after the exam) -- this guide has hand traces only, no project code.

Topics to study first (order + WHY)

1. **Submit HW2 if still open (through Sat Sep 26 2:00 PM)** -- WHY: nearest graded deadline.

2. **345-E1-t2-asymp + 345-E1-t2-rec -- Big-O witnesses + Master Theorem** -- WHY: HW1 score signal was incomplete proofs; Quiz 1 signal was definitions; Topic 2 is already lectured.
3. **345-E1-t3-basics + 345-E1-t3-search -- graphs, BFS, DFS** -- WHY: current lecture weeks 5-6; high-trace value.
4. **345-E1-t3-paths then t3-mcst-topo** -- WHY: in the Topic 3 slides; study Dijkstra now; MCST/topo **only as far as lectured**.
5. **345-E1-t1-* lists / storage / recursion** -- WHY: Topic 1 already lectured; keep warm after Topic 2/3.

Confirmed vs schedule-based: Date confirmed. Topics 1-3 are **provisional from lectures so far**, not a published exam blueprint. Topic 4 is excluded.

Day-by-day tasks (minutes; no class-block study)

When	Window	Task	Guide / slides / problems	Min
Thu Sep 24	After 6:15 PM (345 class 2:00-3:15 already past)	HW2 late-window if not submitted. Exam study only if HW2 is in.	HW2 on Gradescope	HW2
Fri Sep 25	Anytime	HW2 finish first. Then Topic 1 lists/storage/recursion.	t1-*; P1-P3; TR5	50 after HW2
Sat Sep 26	HW2 late cutoff 2:00 PM	After 2:00: Topic 2 counting + Big-O witnesses. Core session -- write PR1 in all four parts.	t2-count, t2-asymp; P4; PR1, Q3	70
Sun Sep 27	After 252 deep time	Topic 2 Master Theorem.	t2-rec; excerpt Topic 2 p.68; P5; TR1	45
Mon Sep 28 - Thu Oct 1	252 Test 2 outranks	Optional 15-min proof flash only.	PR2 one pass	15
Fri Oct 2	Anytime	Topic 3 basics + BFS/DFS.	t3-basics, t3-search; P7-P10; TR3	80
Sat Oct 3	Anytime	Dijkstra; MCST/topo if lectured .	t3-paths; excerpt Topic 3 p.40; TR4; TR6 only if lectured	70
Sun Oct 4	Light -- 337 mock is today	One proof rewrite.	PR1 or PR3	25
Mon Oct 5	Anytime	Timed mock: Q1-Q8 in 15 min, then PR1-PR3 + TR1-TR4 in 45 min.	Sec 4b	75
Tue Oct 6	337 midterm today -- 345 optional after 6:15 PM (CSC 380 meets 5:00-6:15)	Light flash.	Q5, Q8	15

When	Window	Task	Guide / slides / problems	Min
Wed Oct 7	No class	Light mixed. No cram.	All 10 IDs, light	35
Thu Oct 8 morning	Before 9:30 AM or 10:45-12:30 only . CSC 252 meets 9:30-10:45, and the 335 midterm runs 12:30-1:45, leaving just a 15-minute walk/rest gap before 2:00	Calm buffer. Exam clock NV. Look at the four-part proof template once, then stop.	--	10

Rows before Fri Oct 2 are **bank-ahead** except **Sat Sep 26 (Topic 2 proofs)**, which is core because HW1 feedback pointed at incomplete proofs. **Is this enough time?** The core is about 5.3 hours (Sat Sep 26 plus Fri Oct 2 - Mon Oct 5), and the bank-ahead rows add about 2 more. That is enough for Topics 1-3 **if** the Mon Oct 5 mock happens and misses are redone. If Topic 3 lecture gets further than expected (MCST/topo taught on Oct 1), move 25 minutes from the Sun Oct 4 proof rewrite to TR6.

Final self-check

- ☐ HW2 submitted (late window Sat Sep 26 2:00 PM if needed)
- ☐ Blind pass over all 10 IDs
- ☐ Four-part proof habit written three times (statement, witnesses, inequality, conclusion)
- ☐ BFS / DFS / Dijkstra traces redrawn, with a distance table after each Dijkstra step
- ☐ PR2 ("not $O(n)$ ") written from scratch: the contradiction names a specific n
- ☐ MCST/topo only if lectured
- ☐ Topic 4 still excluded
- ☐ Wednesday light -- no night-before cram
- ☐ Energy after the 335 midterm the same day

Slide excerpts (personal study channel only)

Excerpt 1 -- Master Theorem cases (CSC 345, Topic 2, page 68)

Slide image (CSC 345 Topic 2 p.68 Master Theorem) is not mirrored in this repo. It lives on the Munch box at `/home/box/shared/munch/study-guides/csc345/assessments/excerpts/csc345-topic2-p68-master.png`. The own-words caption below and the original deck page are enough to study from.

In my own words: for $T(n) = a T(n/b) + c n^d$, compare a to b^d . If $a < b^d$ the work outside the recursion wins ($\Theta(n^d)$). If $a = b^d$ the logs appear ($\Theta(n^d \log n)$). If $a > b^d$ the recursion tree wins ($\Theta(n^{\log_b a})$). The slide's running example $T(n) = 2 T(n/2) + n$ has $a=2$, $b=2$, $d=1$ so $a = b^d$ and $T(n)$ is $\Theta(n \log n)$. The Master Theorem does not "solve" the recurrence for a closed form -- it names the growth class. Do not apply it to $T(n-1)$ forms (use unrolling + induction, PR3).

Original deck: `/home/box/shared/munch/study-guides/_meta/d2l-harvest/csc345/csc345-topic2-slides.pdf` (public: <https://professorlynam.github.io/csc345/Topic%202%20Slides.pdf>), page 68. Personal study channel only.

Excerpt 2 -- Dijkstra loop (CSC 345, Topic 3, page 40)

Slide image (CSC 345 Topic 3 p.40 Dijkstra) is not mirrored in this repo. It lives on the Munch box at `/home/box/shared/munch/study-guides/csc345/assessments/excerpts/csc345-topic3-p40-dijkstra.png`. The own-words caption below and the original deck page are enough to study from.

In my own words: start with $\text{dist}(\text{source})=0$ and everyone else infinite. Known begins as $\{\{\text{source}\}\}$; Fringe is the source's neighbors. Each round, finalize the Fringe vertex f with the smallest $d(\text{source}, f)$, move it to Known, add newly reached vertices to Fringe, and relax: if you found a cheaper path through a Known vertex t , set $d(\text{source}, f) = d(\text{source}, t) + w(t, f)$. Do not change a vertex after it is Known. With any negative edge weight, Dijkstra can finalize a vertex too early and return a wrong distance.

Original deck: `/home/box/shared/munch/study-guides/_meta/d2l-harvest/csc345/csc345-topic3-slides.pdf` (public: <https://professorlynam.github.io/csc345/Topic%203%20Slides.pdf>), page 40. Personal study channel only.

Also review: Topic 1 slides (lists / orthogonal lists / recursion) `csc345-topic1-slides.pdf`. Topic 4 is not linked -- skip.

Banner — scope PROVISIONAL: Topics 1-3. The exam scope is **not published**. Practice covers the lectures so far: - **Topic 1, Review:** lists, stacks, queues, linked lists, orthogonal lists, array storage, recursion. - **Topic 2, Algorithm Analysis:** step-counting, profiling, asymptotic notation, recurrences. - **Topic 3, Graphs:** representations, BFS/DFS, shortest paths, and the spanning-tree and topological-sort material in the Topic 3 slides.

Topic 4 (Internal Sort) slides are **not linked yet**, so Topic 4 is excluded. The **date is set:** Thu Oct 8, 2026; no clock time is published.

Canonical maps: `csc345-topic-map.md` · `announced-assessments-inventory-2026-09-24.md` · `site-audit-2026-09-24.md`

1. Header

Field	Value	Status
Course	CSC 345 Analysis of Discrete Structures (FA26), TuTh 2:00–3:15 PM, The Commons 305	Confirmed
Assessment	Exam 1	Confirmed
Date	Thu Oct 8, 2026	Scheduled (syllabus)

Field	Value	Status
Time / format	Exam clock time not published . Format not published ; the syllabus says in person, and Gradescope's role is unknown.	NV
Coverage	Topics 1-3	PROVISIONAL
Topic 3 depth	Topic 3 is scheduled for weeks 5-6 (week 6 has not happened yet). The slides include shortest paths, minimum-cost spanning trees (MCST), and topological sort. How far lecture gets before Oct 8 is unverified.	Watch lecture
Same day	CSC 335 Midterm, 12:30 PM class in the same room (clock time unverified)	Plan energy
Guide updated	2026-09-24 (America/Phoenix) — Opus + Content-fold note	

All times are **Arizona time (MST)**.

Sources

Source	Role
Topic_1_Slides.pdf, Topic_2_Slides.pdf, Topic_3_Slides.pdf (+ txt)	Topic content
csc345_syllabus.pdf	Exam date; quizzes (best 5 count, no make-ups)
Topic map, inventory, site audit	Topic lists per slide deck; provisional label
Gradescope format notes <code>csc345-2026-09-24.md</code>	Homework 1 rubric shape (Questions 1-8); score signals
D2L Content-pass 2026-09-24	Content gap persists (iframe blank). Scope remains public Topics 1-3 only; no D2L Content download.

2. Topic checklist

Order	Section ID	Topic	Blind [x]	Check [x]	Retake [x]
1	345-E1-t1-lists	Stacks, queues, singly/doubly/circular linked lists			
2	345-E1-t1-storage	Array storage (row-/column-major) and orthogonal lists			
3	345-E1-t1-rec	Recursion			
4	345-E1-t2-count	Step-counting and profiling			
5	345-E1-t2-asym	Big-O / Omega / Theta with witnesses			
6	345-E1-t2-rec	Recurrences and the Master Theorem			
7	345-E1-t3-basics	Graph definitions and representations			

Order	Section ID	Topic	Blind [x]	Check [x]	Retake [x]
8	345-E1-t3-search	BFS and DFS			
9	345-E1-t3-paths	Shortest paths (Dijkstra)			
10	345-E1-t3-mcst-topo	MCST (Prim/Kruskal) and topological sort — only as far as lectured			

3. Explanations + worked examples

Topic 1 — Lists, stacks, queues (345-E1-t1-lists)

- A **stack** is LIFO (push and pop at the top). A **queue** is FIFO (enqueue at the rear, dequeue at the front).
- **Singly linked list with only a head:** prepend is $O(1)$, but appending at the end is $O(n)$ because you must walk to the end. A **tail pointer** makes append $O(1)$. Deleting the tail is still $O(n)$ in a singly linked list, because you need the node before it.
- A **doubly linked list** can delete a known node in $O(1)$.
- **Circular list** where `tail.next` is the front. To append `new`: set `new.next = tail.next` (the front), then `tail.next = new`, then `tail = new`.
- Python's `list` is a dynamic **array**, not a linked list (the slides make this point).

Topic 1 — Array storage and orthogonal lists (345-E1-t1-storage)

- A 2D array with R rows and C columns, element size s , and 0-based indices:
- **Row-major:** $\text{addr}(i, j) = \text{base} + (i \cdot C + j) \cdot s$
- **Column-major:** $\text{addr}(i, j) = \text{base} + (j \cdot R + i) \cdot s$
- An **orthogonal list** stores a sparse matrix as nodes for the nonzero entries only. Each node has (row, col, value) plus a **right** link (next nonzero in its row) and a **down** link (next nonzero in its column), with row and column header lists. Memory is $O(R + C + \text{nonzeros})$ instead of $O(R \cdot C)$.
- PP1 used this structure. This guide gives concepts only, with no project code.

Topic 1 — Recursion (345-E1-t1-rec)

- A recursive definition needs a **base case** and a **recursive case** that moves toward the base.
- To trace a recursive call, draw the call stack. Each call waits for its sub-calls to return.

Topic 2 — Step-counting and profiling (345-E1-t2-count)

- The **instance characteristic** is the input size, n .
- **Step-counting** counts basic operations as a function of n . **Profiling** measures real run time. Profiling needs working code and is noisy (other processes, caching, hardware), so the slides prefer counting and asymptotics for comparisons.

- **Nested loop pattern:** for $i = 1..n$: for $j = 1..i$: step does $1 + 2 + \dots + n = n(n+1)/2$ steps, which is $\Theta(n^2)$.

Topic 2 — Asymptotic notation (345-E1-t2-asym)

- **f is $O(g)$** if there exist $c > 0$ and n_0 such that $f(n) \leq c \cdot g(n)$ for all $n \geq n_0$. Write "f is $O(g)$ " or "f in $O(g)$ "; the slides warn against " $f = O(g)$ ".
- **Omega** is the lower-bound version ($f(n) \geq c \cdot g(n)$). **Theta** means both O and Ω .
- **A complete proof has four parts:** (1) state what you are proving; (2) choose c and n_0 explicitly; (3) show the inequality holds for every $n \geq n_0$; (4) conclude. Homework 1 lost points that, per the score signal, likely relate to proof completeness. The rubric comments were not captured, so this is a likely cause, not a confirmed one.
- **Worked example:** $7n + 8$ is $O(n)$. For $n \geq 1$, $7n + 8 \leq 7n + 8n = 15n$, so $c = 15$ and $n_0 = 1$.
- **Showing something is not $O(g)$:** assume c and n_0 exist, then exhibit an $n \geq n_0$ that breaks the inequality.

Topic 2 — Recurrences and the Master Theorem (345-E1-t2-rec)

For $T(n) = a \cdot T(n/b) + c \cdot n^d$ with $a \geq 1$, $b > 1$, $d \geq 0$ (the form used in the slides): - If $a < b^d$, then $T(n)$ is $\Theta(n^d)$. - If $a = b^d$, then $T(n)$ is $\Theta(n^d \log n)$. - If $a > b^d$, then $T(n)$ is $\Theta(n^{\log_b a})$.

Worked example: $T(n) = 2T(n/2) + n$ has $a = 2$, $b = 2$, $d = 1$. Since $a = b^d$, $T(n)$ is $\Theta(n \log n)$.

Recurrences that do not match this form, such as $T(n) = T(n-1) + c$, can be solved by **unrolling**, and the result can be proved by **induction**.

Topic 3 — Graph basics (345-E1-t3-basics)

- $G = (V, E)$. Undirected edges are sets $\{u, v\}$; directed edges are ordered pairs (u, v) .
- Key terms: adjacent, degree (for directed graphs, in-degree and out-degree), path, **simple path** (no repeated vertices), cycle, connected, DAG (directed acyclic graph).
- **Handshake lemma:** in an undirected graph, $\sum \deg(v) = 2|E|$.

Representation	Space	Is (u, v) an edge?	List the neighbors of u
Adjacency matrix	$\Theta(V^2)$	$O(1)$	$O(V)$
Adjacency list	$\Theta(V + E)$	$O(\deg u)$	$O(\deg u)$

Topic 3 — BFS and DFS (345-E1-t3-search)

- **BFS** uses a queue and visits vertices in layers by edge-distance. On an unweighted graph it finds shortest paths by edge count.
- **DFS** uses a stack or recursion and goes deep first.
- Both run in $O(V + E)$ with adjacency lists, or $O(V^2)$ with a matrix.
- If the graph is disconnected, restart the search from any vertex not yet visited.

- **Tie-break convention:** when the problem says "alphabetical order", always take neighbors in that order.

Topic 3 — Shortest paths (345-E1-t3-paths)

Dijkstra's algorithm (single source, non-negative weights): 1. Set $\text{dist}(\text{source}) = 0$ and every other distance to ∞ . 2. Repeatedly **finalize** the unfinalized vertex with the smallest distance. 3. For each edge out of it, **relax**: if $\text{dist}(u) + w < \text{dist}(v)$, set $\text{dist}(v) = \text{dist}(u) + w$.

Dijkstra can give wrong answers when there are negative edge weights, because a finalized vertex is never revisited.

Worked example (small). Directed edges $S \rightarrow A$ 2, $S \rightarrow B$ 5, $A \rightarrow B$ 1. 1. Finalize S (0). Relax: $A = 2$, $B = 5$. 2. Finalize A (2), the smallest unfinalized. Relax $A \rightarrow B$: $2 + 1 = 3 < 5$, so $B = 3$. 3. Finalize B (3).

Final distances: S 0, A 2, B 3. Write the distance table after every finalize step; graders usually look for it. TR4 has the same shape with one more vertex.

Topic 3 — MCST and topological sort (345-E1-t3-mcst-topo) — only as far as lectured

- **Prim:** grow one tree, adding the cheapest edge that leaves the tree each time.
- **Kruskal:** sort all edges by weight, and add each edge unless it would create a cycle (tracked with union-find).
- **Topological sort** orders a DAG so every edge goes from earlier to later. One method: repeatedly remove a vertex with in-degree 0. Another: order vertices by reverse DFS finish time.
- PP2 (due Oct 13) implements graph algorithms. This guide gives **hand traces only**, with no project code.

4. Practice questions (core)

P1 (recall). Which structure is FIFO, and which is LIFO?

P2 (solve). In a singly linked list with only a head pointer, what are the costs of appending at the end and of prepending?

P3 (trace). A circular list's tail points at the last node. Give the pointer steps to append X .

P4 (solve). Is $3n^2 + 2n$ in $O(n^2)$? Justify with c and n_0 .

P5 (Master). Solve $T(n) = 4T(n/2) + n$ with the Master Theorem. State a , b , d , and the case.

P6 (recall). Why is wall-clock timing a weak sole measure of an algorithm?

P7 (define). Give V and E for the undirected square $A-B-C-D-A$.

P8 (BFS). Edges A-B, B-C, A-D. What is the BFS order from A, taking neighbors alphabetically?

P9 (DFS). Same graph. What is a DFS order from A, recursive, taking neighbors alphabetically?

P10 (explain). Which data structure drives BFS, and which drives DFS?

P11 (recall). When is Exam 1?

P12 (explain). Why is Topic 4 excluded from this guide?

4b. Gradescope-format practice — Practice recommendation

Where this format comes from. The Gradescope format notes show Homework 1 graded as rubric Questions 1-8 (plus a 0-point late-adjustment item) on a written PDF, and a Quiz 1 scored out of 10. The syllabus says quizzes are short, the best five count, and there are no make-ups. The **Exam 1 format is not published**. These items mirror the quiz grain and homework proof habit. They are not a claimed exam format.

Score signals, in words only (policy kept even though this copy now lives in the private `begoneBOT/bob-jeff` repo): the Quiz 1 score suggests tightening **definitions**, and the Homework 1 deductions suggest writing **complete proofs**.

Timed mock (Mon Oct 5): do Q1-Q8 in 15 minutes, then PR1-PR3 and TR1-TR4 in 45 minutes.

Quiz-grain concept checks (about 10 points total)

GS-345-Q1. In one sentence each, give the removal order of a stack and of a queue.

GS-345-Q2. True or false (fix it if false): "Python's `list` is a linked list."

GS-345-Q3. Give witnesses c and n_0 showing that $2n + 5$ is $O(n)$.

GS-345-Q4. For $T(n) = 2T(n/2) + n$, state a , b , d , and the Master case.

GS-345-Q5. Name the frontier data structure for BFS and for DFS.

GS-345-Q6. What is the difference between a path and a simple path?

GS-345-Q7. A 4×5 int array (4 bytes per element) is stored row-major at base 1000. What is the address of $A[2][3]$?

GS-345-Q8. Give the space cost of an adjacency matrix and of an adjacency list, in terms of V and E .

Proof write-ups (Homework-rubric style — write each as one complete answer)

GS-345-PR1. Prove that $5n^2 + 3n$ is $O(n^2)$. Use all four parts: statement, witnesses, inequality for all $n \geq n_0$, conclusion.

GS-345-PR2. Prove that n^2 is **not** $O(n)$.

GS-345-PR3 (induction habit). Let $T(1) = 1$ and $T(n) = T(n-1) + 2$ for $n > 1$. Prove by induction that $T(n) = 2n - 1$ for all $n \geq 1$.

GS-345-PR4. Prove the handshake lemma: $\sum \deg(v) = 2|E|$ for any undirected graph.

Traces and short solves

GS-345-TR1 (Master x3). Solve each: (a) $T(n) = 8T(n/2) + n^2$; (b) $T(n) = T(n/2) + 1$; (c) $T(n) = 2T(n/4) + n$.

GS-345-TR2 (step count). Count exactly how many times `x++` runs, as a function of n :

```
for i = 1 to n
  for j = 1 to i
    x++
```

GS-345-TR3 (BFS/DFS with distances). The undirected edges are A-B, A-C, B-D, C-D, D-E, E-F. Take neighbors alphabetically. - (a) Give the BFS order from A and each vertex's BFS distance. - (b) Give the recursive DFS order from A.

GS-345-TR4 (Dijkstra). The directed weighted edges are $S \rightarrow A$ 4, $S \rightarrow B$ 1, $B \rightarrow A$ 2, $A \rightarrow C$ 1, $B \rightarrow C$ 5. Run Dijkstra from S. Give the order in which vertices are finalized and the final distances.

GS-345-TR5 (stack/queue trace). On an empty stack, run push 1, push 2, pop, push 3, push 4, pop, pop. What is popped, and what remains? Repeat with a queue (enqueue and dequeue).

GS-345-TR6 (only if lectured: Kruskal / topological sort). - (a) Undirected weighted edges: A-B 3, A-C 1, B-C 2, B-D 4, C-D 5. List the edges Kruskal adds and the total weight. - (b) DAG edges: $A \rightarrow B$, $A \rightarrow C$, $B \rightarrow D$, $C \rightarrow D$. Give one valid topological order.

5. ANSWER KEY

Core keys

A1. A queue is FIFO; a stack is LIFO.

A2. Prepend is $O(1)$. Appending at the end is $O(n)$ without a tail pointer, because you must walk the whole list.

A3. $X.next = tail.next$ (the front), then $tail.next = X$, then $tail = X$. The order matters: doing $tail.next = X$ first loses the front. Edge case: if the list is empty ($tail == null$), set $X.next = X$ and $tail = X$.

A4. Yes. For $n \geq 1$, $3n^2 + 2n \leq 3n^2 + 2n^2 = 5n^2$, so $c = 5$ and $n_0 = 1$.

A5. $a = 4$, $b = 2$, $d = 1$, so $b^d = 2 < 4 = a$. This is the $a > b^d$ case: $\Theta(n^{(\log_2 4)}) = \Theta(n^2)$.

A6. It requires finished, debugged code. It mixes in noise from the operating system, caching, and hardware. And it measures one machine and one input, not growth rate. Operation counting and asymptotics avoid all three problems.

A7. $V = \{A, B, C, D\}$. $E = \{\{A,B\}, \{B,C\}, \{C,D\}, \{D,A\}\}$.

A8. A, B, D, C.

A9. A, B, C, D. From A go to B, then C; back up to A; then visit D.

A10. BFS uses a queue. DFS uses a stack (or the call stack when recursive).

A11. Thu Oct 8, 2026. No clock time is published.

A12. Topic 4 slides are not linked yet, and the provisional scope label is Topics 1-3.

Gradescope-format keys

GS-345-Q1. A stack removes the most recently added item first (LIFO). A queue removes the oldest item first (FIFO).

GS-345-Q2. False. It is a dynamic array.

GS-345-Q3. For $n \geq 1$, $2n + 5 \leq 2n + 5n = 7n$, so $c = 7$ and $n_0 = 1$.

GS-345-Q4. $a = 2$, $b = 2$, $d = 1$. Since $a = b^d$, $T(n)$ is $\Theta(n \log n)$.

GS-345-Q5. BFS uses a queue. DFS uses a stack (or recursion).

GS-345-Q6. A simple path never repeats a vertex. A general path may repeat vertices, unless the course definition restricts it.

GS-345-Q7. $1000 + (2 \cdot 5 + 3) \cdot 4 = 1000 + 52 = \mathbf{1052}$.

GS-345-Q8. Matrix: $\Theta(V^2)$. List: $\Theta(V + E)$.

GS-345-PR1. 1. **Claim:** there exist c, n_0 with $5n^2 + 3n \leq c \cdot n^2$ for all $n \geq n_0$. 2. **Witnesses:** $c = 8, n_0 = 1$. 3. **Inequality:** for $n \geq 1$ we have $n \leq n^2$, so $3n \leq 3n^2$, and therefore $5n^2 + 3n \leq 5n^2 + 3n^2 = 8n^2$. 4. **Conclusion:** $5n^2 + 3n$ is $O(n^2)$.

GS-345-PR2. Suppose $c > 0$ and n_0 exist with $n^2 \leq c \cdot n$ for all $n \geq n_0$. Dividing by $n > 0$ gives $n \leq c$ for all $n \geq n_0$. Take $n = \max(n_0, \text{ceil}(c) + 1)$. Then $n \geq n_0$ but $n > c$, a contradiction. So n^2 is not $O(n)$.

GS-345-PR3. - **Base case:** $T(1) = 1 = 2 \cdot 1 - 1$. [x] - **Inductive hypothesis:** $T(k) = 2k - 1$ for some $k \geq 1$. - **Inductive step:** $T(k+1) = T(k) + 2 = (2k - 1) + 2 = 2(k+1) - 1$. [x] - **Conclusion:** by induction, $T(n) = 2n - 1$ for all $n \geq 1$.

GS-345-PR4. Each edge $\{u, v\}$ adds exactly 1 to $\deg(u)$ and 1 to $\deg(v)$, so each edge contributes exactly 2 to the total sum $\deg(v)$. Summing over all edges gives $2|E|$.

GS-345-TR1. - (a) $a = 8, b = 2, d = 2$; $b^d = 4 < 8$, so $\Theta(n^{(\log_2 8)}) = \Theta(n^3)$. - (b) $a = 1, b = 2, d = 0$; $b^d = 1 = a$, so $\Theta(n^{a \log b}) = \Theta(\log n)$. - (c) $a = 2, b = 4, d = 1$; $b^d = 4 > 2$, so $\Theta(n)$.

GS-345-TR2. $1 + 2 + \dots + n = \frac{n(n+1)}{2}$, which is $\Theta(n^2)$.

GS-345-TR3. - (a) BFS order: A, B, C, D, E, F. Distances: A 0, B 1, C 1, D 2, E 3, F 4. - (b) DFS order: A, B, D, C, E, F. From A go to B, then D. D's first unvisited neighbor is C, whose neighbors are all visited, so back up to D. Then visit E, then F.

GS-345-TR4. - Finalized order: S (0), B (1), A (3, via B: $1 + 2$ beats the direct 4), C (4, via A: $3 + 1$ beats $1 + 5 = 6$). - Final distances: S 0, B 1, A 3, C 4.

GS-345-TR5. - Stack: pops 2, 4, 3; remaining [1]. - Queue: dequeues 1, 2, 3; remaining [4].

GS-345-TR6. - (a) Kruskal considers A-C 1 (add), B-C 2 (add), A-B 3 (skip, it would create a cycle), then B-D 4 (add). The total weight is $1 + 2 + 4 = 7$. - (b) A, B, C, D (A, C, B, D is also valid).

6. Common mistakes + readiness

- Saying Python's `list` is a linked list.
- Forgetting the tail update on a circular append.
- Writing " $f = O(g)$ ", or skipping explicit witnesses c and n_0 .
- Proofs missing a part, especially the conclusion line or " $\text{for all } n \geq n_0$ ".
- Misreading a, b , or d in the Master Theorem, or applying it to $T(n-1)$ forms.
- Running BFS or DFS on a disconnected graph without restarting.
- Mixing up row-major and column-major address formulas.
- Relaxing an edge in Dijkstra after its target is already finalized.

Readiness: a blind pass over all 10 IDs, the four-part proof habit written three times, redrawn BFS/DFS/Dijkstra traces, a light mixed review on Wednesday, and no cram the night before.

Start Here (top of this guide) is the canonical day-by-day. The later prep table is kept as a short copy.

7. Day-by-day prep

Day	Focus	IDs
Fri Sep 25	Topic 1: lists, storage, recursion	t1-*
Sat Sep 26	Topic 2: counting, asymptotics, proofs	t2-count, t2-asym
Sun Sep 27	Topic 2: recurrences, Master Theorem	t2-rec
Fri Oct 2	Topic 3: basics, BFS, DFS	t3-basics, t3-search
Sat Oct 3	Topic 3: Dijkstra, plus MCST/topological sort if lectured	t3-paths, t3-mcst-topo
Mon Oct 5	Timed mock (\$4b), then even rotation	all
Tue Oct 6	Optional light flash only (the 337 midterm is today)	—
Wed Oct 7	Light mixed review, not a cram	all
Thu Oct 8 morning	Calm buffer	—

The Tuesday/Thursday class days of Sep 29 and Oct 1 are left light. PP2 is due Tue Oct 13, after the exam.

8. Changes in this review (2026-09-24)

- **Filled Topic 1-3 gaps** from the site audit and topic map: orthogonal lists and array storage, recursion, step-counting vs profiling, Omega/Theta, shortest paths (Dijkstra), and MCST/topological sort (marked "only as far as lectured").
- Added four proof write-ups (PR1–PR4), including a "not $O(n)$ " proof and an induction proof, since Homework 1 was induction proofs. Added traces TR1–TR6. All have keys.
- Restored the **syllabus** facts "best 5 quizzes count, no make-ups". These come from the syllabus, not Gradescope.
- Fixed the ambiguous "Fri Sep 25 / Oct 2" plan row. Every row now has one date.
- Described scores in words only (policy kept; the repo is now private `begoneBOT/bob-jeff`).
- Changed times to Arizona time.

8c. Opus 5.5 review (2026-09-24 evening, America/Phoenix)

- Re-traced every key: BFS/DFS orders and distances (P8, P9, TR3), Dijkstra (TR4), Kruskal and topological sort (TR6), the stack/queue trace (TR5), all Master Theorem cases (P5, Q4, TR1), the row-major address (Q7), and proofs PR1–PR4. **No wrong answers found.**
- **Class-block fix:** the Thu Oct 8 buffer said "before 2:00 PM (and after 335 midterm)", which spans CSC 252 (9:30–10:45) and the 335 midterm (12:30–1:45). It now names the real windows: before 9:30 or 10:45–12:30.

- **Class-block fix:** Tue Oct 6 "after 4:45 PM" is now after 6:15 PM, since CSC 380 meets 5:00-6:15.
- Marked Sat Sep 26 Topic 2 proofs as a core session and the rest of the pre-Oct 2 rows as bank-ahead. Added an "is this enough time?" check.
- Added a small Dijkstra worked example with per-step relaxation, a circular-list empty-list edge case, and a more precise negative-weight statement.
- Removed the "public repository" wording. This copy is in the private repo; the words-only score policy is kept.

9. Disclaimer

The Gradescope format notes describe Homework 1 and Quiz 1 only. The Exam 1 blueprint is **not published**, so §4b is a practice recommendation.

Study aid only — not for submitted work. This guide contains no PP2 code. Follow CSC 345 and UA policies.